

LECTURE 13 (OCTOBER 28TH)

Recall. Let $f(t) \in K[t]$ and E be the splitting field for $f(t)$. We have seen (1) $[E : K]$ and (2) the number of K -embeddings $E \rightarrow \bar{K}$. Today's goal is to study (2) and separability.

CHAPTER 14.4 SEPARABLE EXTENSIONS

Definition. (14.9) Let K be a field. A polynomial $f(t) \in K[t]$ is separable if its factorisation over $\bar{K}$ has no multiple roots. Let F/K be an algebraic extension. An element $\alpha \in F$ is separable over K if the minimal polynomial $\text{irr}(\alpha, K)$ over K is separable. We say that F/K is separable if every $\alpha \in F$ is separable over K .

Proposition. Let $f(t) \in K[t]$ be a nonzero polynomial. Then, $f(t)$ is separable over K if $f(t)$ has no multiple roots in $\bar{K}$. The following are equivalent:

- (i) $f(t)$ is separable over K
- (ii) $f(t)$ and $f'(t)$ have no common root in $\bar{K}$
- (iii) $\gcd(f(t), f'(t)) = 1$

Proof. The proof is yours! □

Proposition. (14.23) Let K be a field of characteristic zero and let $f(t) \in K[t]$ be an irreducible polynomial. Then, $f(t)$ is separable. Moreover, algebraic extensions of fields of characteristic zero are separable.

Proof. Set $f(t) = a_n t^n + \dots + a_1 t + a_0$ so that $f'(t) = n a_n t^{n-1} + \dots + a_1$. Note that $\deg(f') < \deg(f)$ since $\text{char}(K) = 0$. Suppose that $\alpha \in \bar{K}$ is a common root of $f(t)$ and $f'(t)$. Then $\text{irr}(\alpha, K) \mid f'(t), f(t)$. Since $f(t)$ is irreducible over K , $\deg(\text{irr}(\alpha, K)) = \deg(f(t))$. This contradicts to the observation that $\text{irr}(\alpha, K) \mid f'(t)$ because $\deg(f') < \deg(f)$. □

Proposition. (14.24) (Primitive element theorem) Every finite separable extension is simple.

Proof. The proof is ours and is technical. □

Corollary. (14.25) Let F/K be a finite separable extension. Then there are exactly $[F : K]$ number of K -embeddings $F \rightarrow \bar{K}$.

Proof. By the primitive element theorem, $F = K(\alpha)$ for some $\alpha \in F$. Since α is separable over K , $\text{irr}(\alpha, K)$ has no multiple roots in $\bar{K}$. Applying proposition (14.3), the number of K -embeddings $F = K(\alpha) \hookrightarrow \bar{K}$ equals the number of distinct roots in $\bar{K}$ of $\text{irr}(\alpha, K)$ which is equal to

$$\deg(\text{irr}(\alpha, K)) = [K(\alpha) : K]$$

□

Remark. Let $\alpha \in F$ be an algebraic element over K . Then the number of K -embeddings $K(\alpha) \hookrightarrow \bar{K}$ is the number of distinct zeros of $\text{irr}(\alpha, K)$ in $\bar{K}$, and both are less or equal to $\deg(\text{irr}(\alpha, K)) = [K(\alpha), K]$. According to corollary (14.25), the equality holds if and only if α is separable over K .

Definition. (14.26) The separability degree of an algebraic extension F/K denoted $[F : K]_s$, is the number of K -embeddings $F \hookrightarrow \bar{K}$.

LECTURE 14 (OCTOBER 30TH)

Remark. Last class, we learned about separable extensions, and today we will be learning about separability degree.

Definition. (14.9) We called a $\alpha \in F$ separable if $\text{irr}(\alpha, K)$ has no repeated roots in $\bar{K}$.

Proposition. (14.23) If $\text{char}(K) = 0$, every irreducible polynomial in $K[x]$ is separable.

Proposition. (14.24) (Prime element theorem) Finite separable extensions are simple.

Corollary. (14.25) For a finite separable extension F/K , the number of K -embeddings $F \rightarrow \bar{K}$ is $[F : K]$. In fact, for an algebraic $\alpha \in F$ over K ,

$$|\{F \xrightarrow{\sigma} \bar{K} \mid \sigma|_K = 1_K\}| = |\{\beta \in \bar{K} \mid (\text{irr}(\alpha, K))(\beta) = 0\}| \leq \deg(\text{irr}(\alpha, K)) = [K(\alpha), K]$$

Definition. (14.26) Let F/K be an algebraic extension. Then

$$[F : K]_s = |\{\sigma : F \rightarrow \bar{K} \mid \sigma|_K = 1_K\}|$$

where $[F : K]_s$ denotes the separability degree of F/K .

Remark. Definition (14.26) is independent of the choice of the algebraic closure of $\bar{K}$.

Proposition. (14.28) If F/E and E/K are algebraic extensions then

$$[F : K]_s = [F : E]_s [E : K]_s$$

Proof. Proof will be further inquired in the supplementary material. $\square$

Theorem. Let F/K be a finite extension. Then F/K is separable if and only if $[F : K]_s = [F : K]$.

Remark. In the case where $F = K(\alpha)$, we already have observed the result.

Remark. For a finite extension F/K ,

$$[F : K]_s \leq [F : K]$$

Proof. (if) Let $\alpha \in F$. Consider the tower $F/K(\alpha)/K$. Note that

$$\begin{aligned} [F : K] &= [F : K(\alpha)][K(\alpha) : K] \\ [F : K]_s &= [F : K(\alpha)]_s [K(\alpha) : K]_s \end{aligned}$$

and $[F : K]$ implies that, through the remark, $[F : K(\alpha)] = [F : K(\alpha)]_s$ and $[K(\alpha) : K] = [K(\alpha) : K]_s$. That is, α is separable over K .

(only if) Ours! $\square$

Corollary. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be elements of $\bar{K}$. Then $\alpha_1, \dots, \alpha_n$ are separable over K if and only if $K(\alpha_1, \dots, \alpha_n)$ is separable over K .

Corollary. Let F/E and E/K be finite extensions. F/K is separable if and only if F/E and E/K are separable.

Proof. To prove this, go back to an analogous statement for algebraic extensions. $\square$

Corollary. Let F/K be an algebraic extension then the set of separable elements of F over K forms a subfield of F .

LECTURE 15 (NOVEMBER 4TH)

Remark. Last time, we saw various properties of separable extensions. Today, we will learn about normal extensions.

Proposition. (14.23) If $\text{char}(K) = 0$, then every algebraic extension of K is separable.

Proposition. (14.24) (Primitive element theorem) Every finite separable extension is simple.

Definition. (14.26) Let F/K be an algebraic extension. $[F : K]_s$ denotes the separability degree of F/K which is equal to the number of K -embeddings from $F \rightarrow \bar{K}$.

Proposition. (14.28) Let $F/E/K$ be algebraic extensions.

$$[F : K]_s = [F : E]_s [E : K]_s$$

Corollary. If F/K is finite, $[F : K]_s < \infty$ (and $[F : K]_s$ divides $[F : K]$).

CHAPTER 14.3 NORMAL EXTENSIONS

Definition. (14.15) An algebraic extension F/K is normal if for every $\alpha \in F$, the minimal polynomial $\text{irr}(\alpha, K)$ splits in F .

Remark. F/K is normal if and only if the following condition holds: if F contains a zero of an irreducible polynomial $p(t) \in L[t]$, then $p(t)$ splits into linear factors over F .

Remark. (Motivation) Let $p(t) \in K[t]$ be irreducible. Say $p(t) = \prod_{i=1}^n (t - \lambda_i) \in \bar{K}[t]$. Consider a K -embedding $\sigma : E \hookrightarrow \bar{K}$ where $E = K(\lambda_1, \dots, \lambda_n)$ is the splitting field of $p(t)$. We want to figure out the image $\sigma(E) \subset \bar{K}$. We claim that $\sigma(E) = E$. For this, observe that the image of λ_i under σ is α_j for some j . In other words, σ permutes $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ ($\sigma : \{\lambda_1, \dots, \lambda_n\} \rightarrow \{\lambda_1, \dots, \lambda_n\}$). All in all, σ gives rise to a K -automorphism on E when dealing with field extensions in which irreducible polynomials in it split.

Theorem. Let F/K be an algebraic extension. Fix an algebraic closure $\bar{K}$ of K with $K \subset F \subset \bar{K}$. The following are equivalent:

- (i) There exists a subset $S \subset K[t]$ such that F is a splitting field for S .
- (ii) If $\sigma : F \rightarrow \bar{K}$ is a K -embedding, then $\sigma(F) = F$.
- (iii) F/K is normal.

Proof. Assume that condition (ii) holds. To show that F is normal over K , let α be an element of F . Suppose we are given a zero β of $\text{irr}(\alpha, K)$ in $\bar{K}$. We wish to show that $\beta \in F$.

First we have a K -isomorphism $\sigma_0 : K(\alpha) \rightarrow K(\beta)$. Using the isomorphism extension theorem, we can extend σ_0 to a K -embedding $\sigma : F \rightarrow \bar{K}$. By our assumption, $\sigma(F) = F$. In particular, $\beta = \sigma(\alpha) \in F$.

$$\begin{array}{ccc}
F & \xrightarrow{\sigma} & \bar{K} \\
\uparrow & & \uparrow \\
K(\alpha) & \xrightarrow{\sigma_0} & K(\beta) \\
\uparrow & & \uparrow \\
K & \longrightarrow & K
\end{array}$$

□

Lemma. Let F/K be an algebraic extension, and let $\sigma : F \rightarrow F$ be a K -embedding. Then $\sigma(F) = F$.

Proof. Let $\alpha_0 \in F$ be an element. Set $\text{irr}(\alpha_0, K) = (t - \alpha_0)(t - \alpha_1) \dots (t - \alpha_r)(t - \beta_1) \dots (t - \beta_s) \in \bar{K}[t]$ where $\alpha_i \in F$ and $\beta_j \in \bar{K} \setminus F$. Then σ permutes $\{\alpha, \alpha_1, \dots, \alpha_r\}$ so that $\alpha_0 = \sigma(\alpha_i)$ for some $0 \leq i \leq r$. □

Remark. Let F/K be a finite extension. Then F/K is normal if and only if

$$[F : K]_s = |\text{Aut}(F/K)|$$

LECTURE 16 (NOVEMBER 6TH)

Remark. Last class we have learned normal extensions. This class, we learn more about normal extensions and newly finite fields.

Theorem. Let F/K be an algebraic extension and $\bar{K}$ be an algebraic closure of K with $F \subset \bar{K}$. The following are equivalent:

- (i) There exists $S \subset K[t]$ such that F is a splitting field for S (that is, $F = K(S_0)$ where S_0 is the set of zeros of $f(t) \in S$).
- (ii) For each K -embedding $\sigma : F \rightarrow \bar{K}$, $\sigma(F) = F$.
- (iii) F/K is normal. That is, for each $\alpha \in F$, all zeros of $\text{irr}(\alpha, K)$ in $\bar{K}$ lie in F .

Example. (14.13) Consider $p(t) = t^4 + t^3 + t^2 + t + 1 \in \mathbf{Q}[t]$. Let $E = \mathbf{Q}(\xi)$, ξ is a primitive 5th root of unity. Observe that σ_i can be defined to send any $\xi \mapsto \xi^i$ for $1 \leq i \leq 4$.

$$\begin{array}{ccc}
& & \bar{Q} \\
& & \uparrow \\
E = Q(\xi) & \xrightarrow{\sigma_i} & Q(\xi^i) = Q(\xi) \\
\uparrow & & \uparrow \\
Q & \xrightarrow{\quad} & Q
\end{array}$$

Now consider $p(t) = t^3 - 2 \in Q[t]$. Let $E = Q(2^{1/3}, \xi)$ where ξ is a primitive 3rd root of unity.

$$\begin{array}{ccc}
F = Q(2^{1/3}) & \xrightarrow{\sigma} & \bar{Q} \\
\uparrow & & \uparrow \\
Q & \xrightarrow{\quad} & Q
\end{array}$$

Here, $\sigma(2^{1/3}) = 2^{1/3}\xi \notin Q(2^{1/3})$.

Corollary. Let F/K be a finite extension. Then F is normal over K if and only if

$$[F : K]_s = |\text{Aut}(F/K)|$$

where $\text{Aut}(F/K)$ is the group of all K -isomorphisms $\sigma : F \rightarrow F$, $\sigma \cdot \mu = \sigma \circ \mu$.

Remark. (Observation) Choose an algebraic closure of K with $K \subset F \subset \bar{K}$. There is an injection

$$\text{Aut}(F/K) \rightarrow \{\sigma : F \rightarrow \bar{K}\}$$

where σ is a K -embedding. Schematically, we have a map

$$\left(\begin{array}{ccc} F & \xrightarrow{\sigma} & F \\ & \searrow & \nearrow \\ & K & \end{array} \right) \mapsto \left(\begin{array}{ccccc} F & \xrightarrow{\sigma} & F & \xrightarrow{\iota} & \bar{K} \\ & \searrow & \downarrow & \nearrow & \\ & & K & & \end{array} \right)$$

Proposition. If F/K is a field extension of degree 2, then F/K is normal over K .

Proof. Choose an element $\alpha \in F \setminus K$. Then $F = K(\alpha)$. Let $\text{irr}(\alpha, K) = t^2 + at + b \in K[t]$. Note that

$$t^2 + at + b = (t - \alpha)(t + (a + \alpha)) \in \bar{K}[t]$$

We claim that F is a splitting field for $\text{irr}(\alpha, K)$. This follows from the observation that $-(a + \alpha)$ and α belong in F . An observation is that for $F/E/K$, if F/K is normal, then F/E is normal. $\square$

CHAPTER 15. GALOIS THEORY

Definition. (15.1) Let F/K be a field extension. Then the Galois group of F/K is the group of K -automorphisms on F . We will denote it by $\text{Gal}(F/K)$.

LECTURE 17 (NOVEMBER 12TH)

Theorem. Consider F/K , an algebraic extension. The following are equivalent:

- (i) $\exists S \subset K[t]$ such that F is a splitting field for S
- (ii) If $\sigma : F \rightarrow \bar{K}$ is a K -embedding, then $\sigma(F) = F$
- (iii) F/K is normal

Corollary. Let F/K be a finite extension. F/K is normal if and only if

$$[F : K]_s = |\text{Aut}(F/K)|$$

Proposition. If $[F : K] = 2$, then F/K is normal.

CHAPTER 15. GALOIS THEORY

15.1 GALOIS GROUPS AND GALOIS EXTENSIONS

Definition. Let F/K be a field extension. The Galois group of F/K , denoted $\text{Gal}(F/K)$ is the group of isomorphisms $\sigma : F \rightarrow F$ that restricts to the identity on K .

Definition. (15.2) Let F/K be an algebraic extension. We will say that F/K is Galois if it is separable and normal.

Proposition. Let F/K be a finite extension. The following are equivalent:

- (i) F/K is Galois
- (ii) $[F : K] = |\text{Aut}(F/K)|$
- (iii) F is a splitting field of a separable polynomial in $K[t]$

Proof. We prove (i) if and only if (ii). Note that $|\text{Aut}(F/K)| \leq [F : K]_s \leq [F : K]$. The first equality holds if and only if F/K is normal and the second holds if and only if F/K is separable.

We now prove (i) to (iii). By the primitive element theorem, we can write $F = K(\alpha)$ for some $\alpha \in F$. Set $f(t) = \text{irr}(\alpha, K) \in K[t]$. Since F is normal over K , F is splitting field of $f(t)$. We note that $f(t)$ is separable since F/K is separable.

Finally, (iii) to (i). Suppose that $F = K(\alpha_1, \dots, \alpha_n)$ is a splitting field of a separable polynomial $f(t)$. Using the normal extension theorem, we see that F is normal over K . Note that each α_i is separable over K . This implies that $K(\alpha_1, \dots, \alpha_n) = F$ is separable. $\square$

Definition. (15.15) Let F/K be a field extension. Let G be a subgroup of $\text{Gal}(F/K)$. The fixed field of G is the subfield:

$$\{\alpha \in F \mid \sigma(\alpha) = \alpha \text{ for all } \sigma \in G\} \subset F$$

denoted F^G .

Remark. (Observation) F^G is indeed a subfield of F containing K ($K \subset F^G \subset F$).

Remark. (Remark) Let $F/E/K$ be field extensions. Then $\text{Gal}(F/E)$ is a subgroup of $\text{Gal}(F/K)$.

Theorem. (15.31 + 15.35) (The main theorem of Galois theory) Let F/K be a finite Galois extension. Then there is an order-reversing bijection between the set of all intermediate fields for F/K and the set of all subgroups of $\text{Gal}(F/K)$.

LECTURE 18 (NOVEMBER 13TH)

Remark. Last time we learned Galois theory. Today, we learn the main theorem and examples of $\text{Gal}(F/K)$.

Definition. (15.2) An algebraic extension F/K is Galois if it is separable and normal.

Remark. A finite extension F/K is Galois if and only if F is splitting field of a separable polynomial in $K[t]$ if and only if

$$[F : K] = |\text{Aut}(F/K)| = |\text{Gal}(F/K)|$$

Remark. Let F/K is finite

$$|\text{Aut}(F/K)| \leq [F : K]_s \leq [F : K]$$

the first equality holds if and only if F/K is normal and the second equality holds if and only if F/K is separable. We see that the two conditions we previously sought were exactly F being normal and separable.

Theorem. (15.31) and (15.35) (Fundamental theorem of Galois theory) Let F/K be a finite Galois extension. Then the Galois correspondence

$$\{E : \text{an intermediate field } K \subset E \subset F\} \longleftrightarrow \{\text{subgroups of } \text{Gal}(F/K)\}$$

where $E \mapsto \text{Gal}(F/E)$ and $H \mapsto F^H$ is a bijection.

Proof. If E is an intermediate field $K \subset E \subset F$, then the extension E/K is Galois (or normal) if and only if $\text{Gal}(F/E)$ is a normal subgroup of $\text{Gal}(F/K)$. In this case, $\text{Gal}(E/K)$ is isomorphic to $\text{Gal}(F/K)/\text{Gal}(F/E)$. $\square$

Example. (15.12) Let $f(t) = t^4 + t^3 + t^2 + t + 1 \in \mathbf{Q}[t]$. Let F = the splitting field of $f(t)$, equal to $\mathbf{Q}(\xi, \xi^2, \xi^3, \xi^4)$. Here, ξ is a primitive 5th root of unity. Note that

(i) $F/\mathbf{Q}$ is finite Galois.

(ii) Let $\sigma \in \text{Gal}(F/\mathbf{Q})$.

$$\begin{array}{ccc} \mathbf{Q}(\xi, \xi^2, \xi^3, \xi^4) & \xrightarrow{\sigma} & \mathbf{Q}(\xi, \xi^2, \xi^3, \xi^4) \\ & \searrow & \swarrow \\ & \mathbf{Q} & \end{array}$$

(ii-i) It is enough to consider $\sigma(\xi)$

(ii-ii) $\sigma(\xi)$ must be zero of $f(t)$ ($f(\xi) = 0$ implies that $f(\sigma(\xi)) = 0$) which implies that $\sigma(\xi) = \xi^i$ for $1 \leq i \leq 4$.

(ii-iii) There exists a $\sigma_i \in \text{Gal}(F/K)$ such that $\sigma_i(\xi) = \xi^i$ for each $1 \leq i \leq 4$ by the isomorphism extension theorem.

(ii-iv) $\text{Gal}(F/\mathbf{Q}) = \{\sigma_1, \sigma_2, \sigma_3, \sigma_4\}$

(ii-v) $(\mathbf{Z}/5\mathbf{Z})^\times \rightarrow \text{Gal}(F/\mathbf{Q})$.

LECTURE 19 (NOVEMBER 18TH)

Remark. Last time, we saw examples of $\text{Gal}(F/K)$. Today, we learn more about the examples and we also learn Galois theory. Note that regarding notation, the textbook uses

$\text{Gal}_K(F)$. However, in our class, we use $\text{Gal}(F/K)$.

Example. (15.11) Let $f(t) = t^4 + t^3 + t^2 + t + 1 \in \mathbf{Q}[t]$, and let F be the splitting field of $f(t)$, $\mathbf{Q}(\xi, \xi^2, \xi^3, \xi^4) = \mathbf{Q}(\xi)$ (where ξ is the primitive 5th root of unity). Calculate $\text{Gal}(F/\mathbf{Q})$.

Proof. (i) For each $n \in \{1, 2, 3, 4\}$, $\sigma_{\bar{n}} : \mathbf{Q}(\xi) \rightarrow \mathbf{Q}(\xi) : \xi \mapsto \xi^n$ is an element of $\text{Gal}(F/\mathbf{Q})$.

(ii) There is a map $\phi : (\mathbf{Z}/5\mathbf{Z})^\times \rightarrow \text{Gal}(F/\mathbf{Q})$ that maps $\bar{n} \mapsto \sigma_{\bar{n}}$. Here, we regard $((\mathbf{Z}/5\mathbf{Z})^\times, +)$ as an abelian group.

(iii) ϕ is an isomorphism and in conclusion, $\text{Gal}(F/\mathbf{Q})$ is a cyclic group of order 4.

Example. (15.13) Now let's turn to $f(t) = t^3 - 2$. Suppose $\sigma \in \text{Gal}(F/\mathbf{Q})$. An important observation is that for each zero $*$ of $f(t)$, $\sigma(*)$ is a zero of $f(t)$ (note that $*$ $\in \{\alpha, \alpha\xi, \alpha\xi^2\}$). Therefore, $\sigma(\alpha) \in \{\alpha, \alpha\xi, \alpha\xi^2\}$. Now we determine $\sigma(\alpha\xi)$. Then

$$\sigma(\alpha\xi) \in \{\alpha, \alpha\xi, \alpha\xi^2\} - \{\sigma(\alpha)\}$$

and also

$$\sigma(\alpha\xi^2) \in \{\alpha, \alpha\xi, \alpha\xi^2\} - \{\sigma(\alpha), \sigma(\alpha\xi)\}$$

The conclusion is that σ restricts to a bijection

$$\{\alpha, \alpha\xi, \alpha\xi^2\} \rightarrow \{\alpha, \alpha\xi, \alpha\xi^2\}$$

Conversely, for each bijection, we can obtain an element of $\text{Gal}(F/\mathbf{Q})$ (since $F/\mathbf{Q}$ is Galois). It turns out that the map $\text{Gal}(F/\mathbf{Q}) \rightarrow S_3$, $\sigma \mapsto (\{\alpha, \alpha\xi, \alpha\xi^2\} \rightarrow \{\alpha, \alpha\xi, \alpha\xi^2\})$ is an isomorphism.

Remark. In example (15.12), we got a group homomorphism

$$\text{Gal}(F/\mathbf{Q}) \rightarrow S_4$$

where

$$\sigma \mapsto (\{\xi, \xi^2, \xi^3, \xi^4\} \rightarrow \{\xi, \xi^2, \xi^3, \xi^4\})$$

Lemma. (15.8) Let F/K be an algebraic extension. Suppose that $F = K(\alpha_1, \dots, \alpha_n)$, where α_i 's are the zeros in F of a polynomial $f(t) \in K[t]$. Then $\text{Gal}(F/K)$ is isomorphic to a subgroup of S_n .

Proof. Each σ determined by the values at α_i . In fact, there is a group homomorphism

$$\text{Gal}(F/K) \rightarrow S_{\{\alpha_1, \dots, \alpha_n\}}$$

Moreover, this map is injective. For this, we will use group actions. $\square$

LECTURE 20 (NOVEMBER 20TH)

Remark. Last class, we have seen the two particular examples $\text{Gal}(\mathbf{Q}(\xi), \mathbf{Q}) \cong \mathbf{Z}/4\mathbf{Z}$ and also $\text{Gal}(\mathbf{Q}(\alpha, \alpha\xi, \alpha\xi^2)/\mathbf{Q}) \cong S_3$. Today, we will be learning Galois theory.

Corollary. (15.8) Let $f(t)$ be a separable polynomial in $K[t]$, and let $F = K(\alpha_1, \dots, \alpha_n)$ be the splitting field of $f(t)$. Then the Galois group $\text{Gal}(F/K)$ is isomorphic to a subgroup of S_n .

Proof. Let $\sigma \in \text{Gal}(F/K)$ be an element. Then for each α_i , its image under σ is a zero of $f(t)$: In fact, σ determines a bijection $\{\alpha_1, \dots, \alpha_n\} \rightarrow \{\alpha_1, \dots, \alpha_n\}$. Moreover, we have a group homomorphism

$$\phi : \text{Gal}(F/K) \rightarrow S_n$$

Since $F = K(\alpha, \dots, \alpha_n)$, ϕ is injective. $\square$

Theorem. (15.31) (Fundamental theorem of Galois theory I) Let F/K be a finite Galois extension. Then the Galois correspondence is a bijection: that is, there is a bijection between the intermediate fields $K \subset E \subset F$ and the subgroups H of $\text{Gal}(F/K)$, given by $E \mapsto \text{Gal}(F/E)$ and $H \mapsto F^H = \{\alpha \in F \mid \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}$.

Proof. We prove only one direction. We first show that the map $E \mapsto \text{Gal}(F/E)$ is injective. For this, it suffices to show that $F^{\text{Gal}(F/E)} = E$ for each subfield $K \subset E \subset F$. For this, observe that $E \subset F^{\text{Gal}(F/E)}$ as for any $\alpha \in E$, $\sigma \in \text{Gal}(F/E)$ satisfies $\sigma(\alpha) = \alpha$, and $\alpha \in F^{\text{Gal}(F/E)}$.

For the other side, note that $F/F^{\text{Gal}(F/E)}$ is a finite Galois extension and that $\text{Gal}(F/F^{\text{Gal}(F/E)})$ is a subgroup of $\text{Gal}(F/E)$. Since $\text{Gal}(F/E)$ is a subgroup of $\text{Gal}(F/F^{\text{Gal}(F/E)})$, we see that $\text{Gal}(F/E) = \text{Gal}(F/F^{\text{Gal}(F/E)})$. To conclude, we observe that

$$[F : F^{\text{Gal}(F/E)}] = |\text{Gal}(F/F^{\text{Gal}(F/E)})| = |\text{Gal}(F/E)| = [F : E]$$

Thus $[F^{\text{Gal}(F/E)} : E] = 1$ and $F^{\text{Gal}(F/E)} = E$.

To prove the surjectivity (of $E \mapsto \text{Gal}(F/E)$), let H be a subgroup of $\text{Gal}(F/K)$. We will show that $H = \text{Gal}(F/F^H)$. For this, let $\alpha \in F$ be an element. Take a maximal subset $\{\sigma_1, \dots, \sigma_r\} \subset H$ such that $\{\sigma_1(\alpha), \dots, \sigma_r(\alpha)\}$ are mutually distinct. We may assume that

$\sigma_1 = 1_F$. We may assume that $\sigma_1 = 1_K$. Let

$$f(t) = \prod_{i=1}^r (t - \sigma_i(\alpha))$$

□

LECTURE 21 (NOVEMBER 25TH)

Remark. Last class, we learned Galois theory and its proof. Today, we will continue the proof and look at some examples.

Theorem. (15.31) (Fundamental theorem of Galois theory I) Let F/K be a finite Galois extension.

$$\begin{aligned} \{K \subset E \subset F\} &\leftrightarrow \{1 \subset H \subset \text{Gal}(F/K)\} \\ E &\mapsto \text{Gal}(F/E) \\ H &\mapsto H \end{aligned}$$

Proof. The injectivity of $E \mapsto \text{Gal}(F/E)$ follows for any intermediate field E . To prove that the map

$$E \mapsto \text{Gal}(F/E)$$

is surjective, let H be a subgroup of $\text{Gal}(F/K)$, we claim that $\text{Gal}(F/F^H) = H$. Let $\alpha \in F$ be an element. Take a maximal subset $\{\sigma_1, \dots, \sigma_r\} \subset H$ such that $\sigma_1(\alpha), \dots, \sigma_r(\alpha)$ are mutually distinct. We may suppose that $\sigma_1 = 1_F$. Let

$$f(t) = \prod_{i=1}^r (t - \sigma_i(\alpha)) \in F[t] \supset F^H[t]$$

Then for each $\tau \in H$, the maximality of the subset implies that

$$\{(\tau \circ \sigma_1)(\alpha), \dots, (\tau \circ \sigma_r)(\alpha)\} \subset \{\sigma_1(\alpha), \dots, \sigma_r(\alpha)\}$$

It then follows that $f(t) \in F^H[t]$. We now complete the proof. Using the fact that $H \triangleleft \text{Gal}(F/F^H)$, $|H| \leq |\text{Gal}(F/F^H)| = [F : F^H]$. Consequently, it will suffice to show that $[F : F^H] \leq |H|$. By the primitive element theorem, $F = F^H(\alpha)$ for some $\alpha \in F$. It follows from the previous argument that

$$[F : F^H] = [F^H(\alpha) : F^H] = \deg(\text{irr}(\alpha, F^H)) \leq f(t) \leq |H|$$

□

Remark. The proof shows the following: for each $\alpha \in F$, let $\sigma_1(\alpha), \dots, \sigma_r(\alpha)$ be the Galois conjugates of α (that is, it is the set $\{\sigma_i(\alpha) \mid \sigma_i \in \text{Gal}(F/K)\}$). Then

$$f(t) = \prod_{i=1}^r (t - \sigma_i(\alpha)) \in F^{\text{Gal}(F/K)}[t] = K[t]$$

Is $f(t) = \text{irr}(\alpha, K)$? According to proposition (15.24), $f(t)$ is indeed the minimal polynomial of α over K .

LECTURE 22 (NOVEMBER 27TH)

Remark. Last class, we have learned the fundamental theorem of Galois theory I. Today, we will look at some examples and the concept of group action.

Theorem. (15.31) For a finite Galois extension F/K , there exists a correspondence between the Galois group (the group of automorphisms) and intermediate fields. Explicitly, the map

$$\{K \subset E \subset F\} \leftrightarrow \{H \triangleleft \text{Gal}(F/K)\}$$

is given by

$$E \mapsto \text{Gal}(F/E)$$

and

$$H \mapsto F^H$$

Definition. (15.22) Let F/K be a finite Galois extension. Let $\alpha \in F$ be an element. The distinct $\sigma(\alpha)$ for $\sigma \in \text{Gal}(F/K)$ are the Galois conjugates of α .

Proposition. (15.24) Let F/K be a finite Galois extension. Let $\alpha \in F$ be an element. Let $\alpha_1, \dots, \alpha_r$ be the Galois conjugates of α . Then

$$f(t) = \prod_{i=1}^r (t - \alpha_i)$$

is the minimal polynomial of α over K .

Proof. It suffices to show that $\deg f(t) = \deg \text{irr}(\alpha, K)$. Note that

$$\deg(\text{irr}(\alpha, K)) = [K(\alpha) : K] = \frac{[F : K]}{[F : K(\alpha)]} = \frac{|\text{Gal}(F/K)|}{|\text{Gal}(F : K(\alpha))|} = [\text{Gal}(F/K) : \text{Gal}(F/K(\alpha))]$$

Consider the action of $\text{Gal}(F/K)$ on the set of the Galois conjugates of α . Applying lemma (12.9), we see that

$$\deg f(t) = r = [\text{Gal}(F/K) : \text{Gal}(F/K(\alpha))]$$

□

Definition. (11.9) (Group actions) Let G be a group and X be a set. An action of G on X is a function $\phi : G \times X \rightarrow X$ such that:

- (i) $\phi(1, x) = x$ for all $x \in X$.
- (ii) $\phi(gh, x) = \phi(g, \phi(h, x))$ for all $g, h \in G$ and all $x \in X$.

Proposition. (11.20) Giving an action of G on X is equivalent to giving a group homomorphism

$$G \rightarrow S_X$$

Proof. Ours! □

Example. (i) G acts on itself by left multiplication $G \times G \rightarrow G : (g, h) \mapsto gh$

(ii) G acts on itself by conjugation $G \times G \rightarrow G : (g, h) \mapsto ghg^{-1}$

Definition. (11.22) Let X be a G -set. The orbit of an element $x \in X$ is the subset $\{g \cdot x \mid g \in G\} \subset X$.

Definition. (12.8) The stabiliser of $x \in X$ is the subgroup

$$G_x = \{g \in G \mid g \cdot x = x\}$$

Proposition. Let S_n act on $\{1, \dots, n\}$. Let $H \triangleleft S_n$. Then is $H = (S_n)_i$ for some i ?

Lemma. (12.9) There is a bijection between the orbit of X and the set G/G_X of left cosets of G_X .

Proof. Define a function $\phi : G_X \rightarrow G/G_X$. Check that it is well defined and that it is bijective. □